

COMPARISON OF VIRTUALBOX & VMWARE WORKSTATION FOR TEST AND DEVELOPMENT ENVIRONMENTS

Module: IT Infrastructure & Automation

Assessment: CA1 (50%)

Student Name: Rakesh Uday Kumar

Student Number: A00047386

Lecturer: Dr. David White

Word Count: 2242 (including ref + appendix section)

1 INTRODUCTION

Virtualization allows to run more than one operating system on a single physical computer. This makes it easier to test, analyse security, and do operational tasks. This report shows how Oracle VirtualBox and VMware Workstation Pro, two of the best desktop virtualization tools, compared to each other in use.

VirtualBox was chosen for its open-source availability, cross-platform support, and ease of setup which makes it great for learning and testing. VMware Workstation Pro was chosen for its strong performance, enterprise reliability, and advanced features such as snapshots, linked clones, and better hardware optimization.

The comparison looks at four main areas: performance and scalability, networking, backup and migration, and security and automation. The test took place in a virtual lab with an Kali linux attacker, a Windows target, and a pfSense firewall that acted as a gateway. The goal is to find out which platform works better, is easier to manage, and is safer for test and development environments.

2 SYSTEM SETUP AND ENVIRONMENT

2.1 HOST SYSTEM

All tests were run on a single physical host to keep results comparable across platforms. The host ran Windows 11 on an Intel Core i7 with 16 GB RAM and a 512 GB SSD. Both Oracle VirtualBox and VMware Workstation Pro 17 were installed on this machine. The global hypervisor settings and resource limits were the same for both programs to make the comparison fair.

Performance:

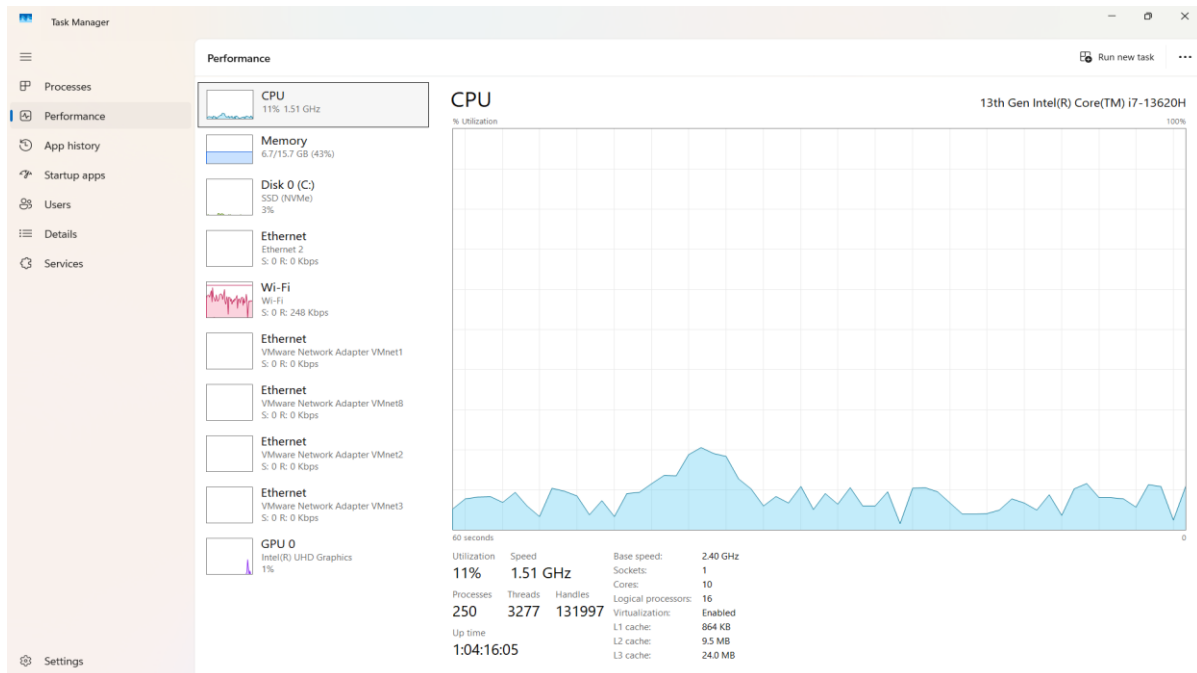


Figure 1: Host Server CPU load.

Memory:

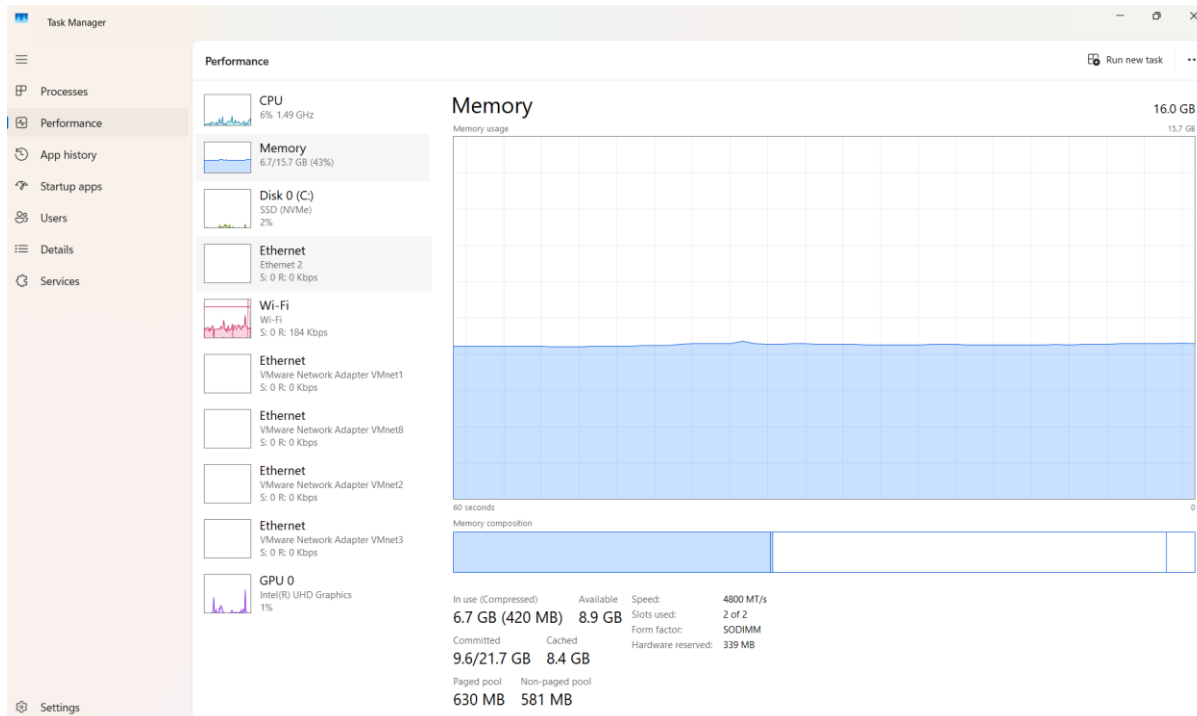


Figure 2: Host Memory details.

2.2 VIRTUAL MACHINES

The lab used three VMs to make a small test-and-development network. The choices for resource allocation were made to find a balance between realism and the amount of space available on the host.

Each ISO was chosen for a specific, practical reason:

- **Kali Linux (Debian)**: was chosen because it is open-source, lightweight, and comes with a lot of cybersecurity and DevOps tools, which makes it perfect for simulating realistic attack behaviour. Its flexible tools (like packet generators, iperf3, fio, sysbench, and others) make it easy to measure CPU, memory, disk, and network performance during tests in a way that can be done repeatedly. These features make Kali linux a useful platform for both controlled performance benchmarking and offensive security exercises.
- **Windows 10**: represents a typical corporate workstation environment and the most common target in enterprise security testing. It has a very different GUI and networking behaviour than Kali Linux, which makes it easier to compare how each hypervisor handles workloads with different operating systems.
- **pfsense**: a free, open-source firewall that provides realistic routing, NAT, and packet-filtering features. pfsense allows to test internal, external, and isolated traffic flows and to monitor or block packets directly.

Considering the host configuration (Intel i7, 16 GB RAM, 512 GB SSD) these VM sizes balance reality and room: pfsense uses minimal resources while Kali Linux and Windows run security tools and services reliably without overloading the host.

VM	Role	CPU	RAM	Disk
pfsense	Firewall	1 vCPU	1 GB	16 GB
Windows 10	Target	2 vCPU	4 GB	40 GB
Kali Linux (Debian)	Attacker	2 vCPU	2 GB	25 GB

Table 1: Specs and provisioning of the VM's

Virtual Box Dashboard:

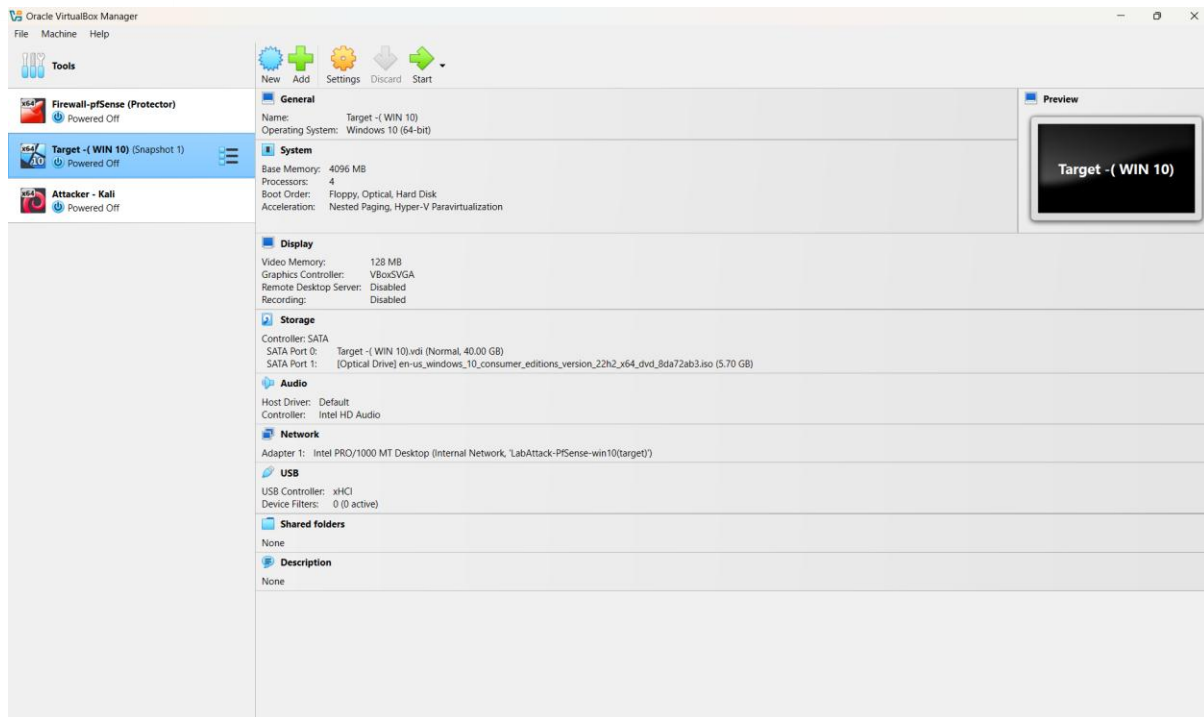


Figure 3: Virtual Box Dashboard

VMware Dashboard:

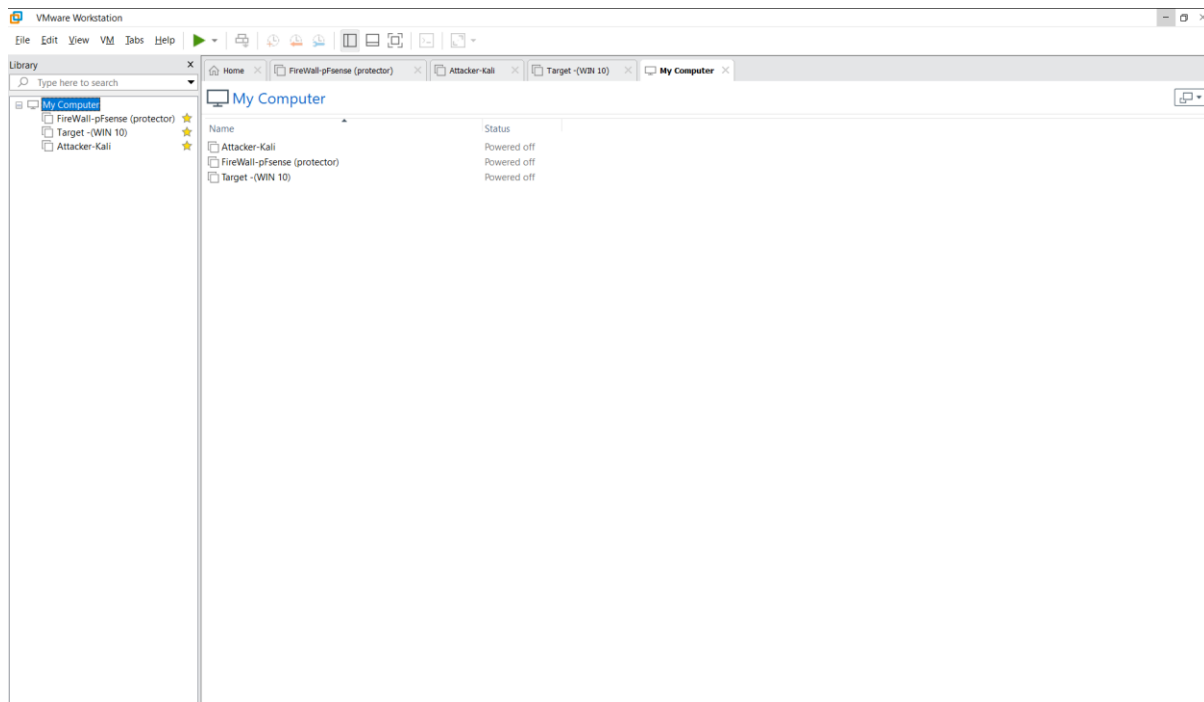


Figure 4: VMware Box Dashboard

3 PERFORMANCE AND SCALABILITY

Each platform was tested for CPU, memory, disk, and network efficiency under identical conditions using tools such as Sysbench, 7-Zip, fio, CrystalDiskMark and iperf3.

3.1 CPU PERFORMANCE

Virtual Box:

- **Kali (Linux Debian):**

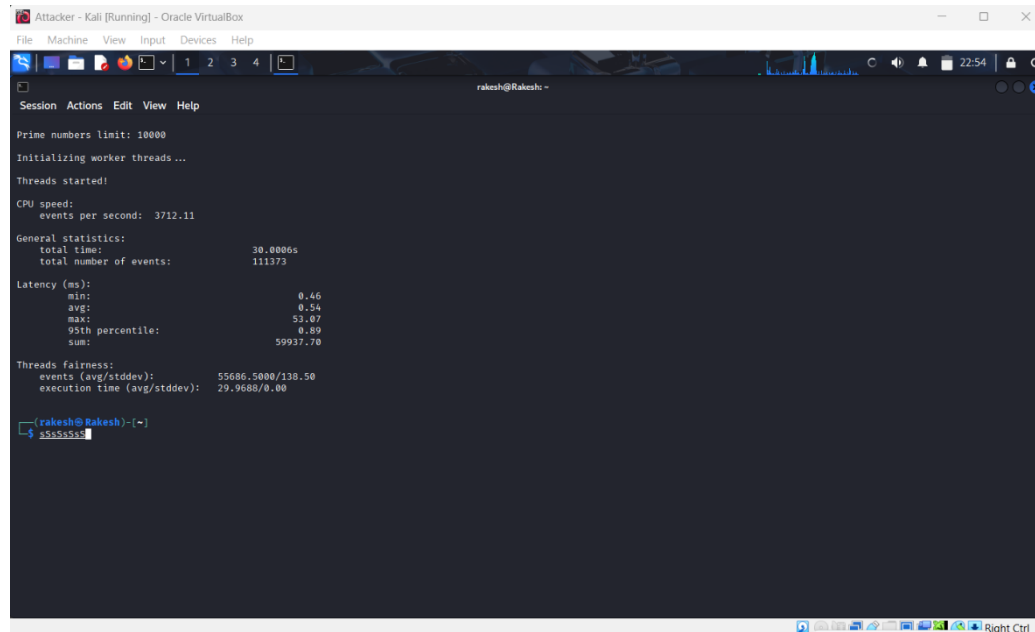


Figure 5: CPU performance VBox (Kali -attacker VM)

- **Windows:**

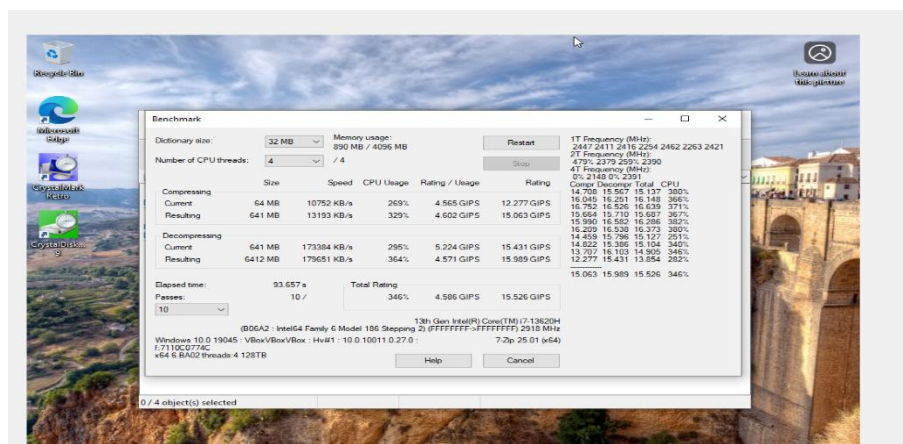


Figure 6: CPU performance VBox (Win 10 - target VM)

VMWare:

- Kali (Linux Debian):**

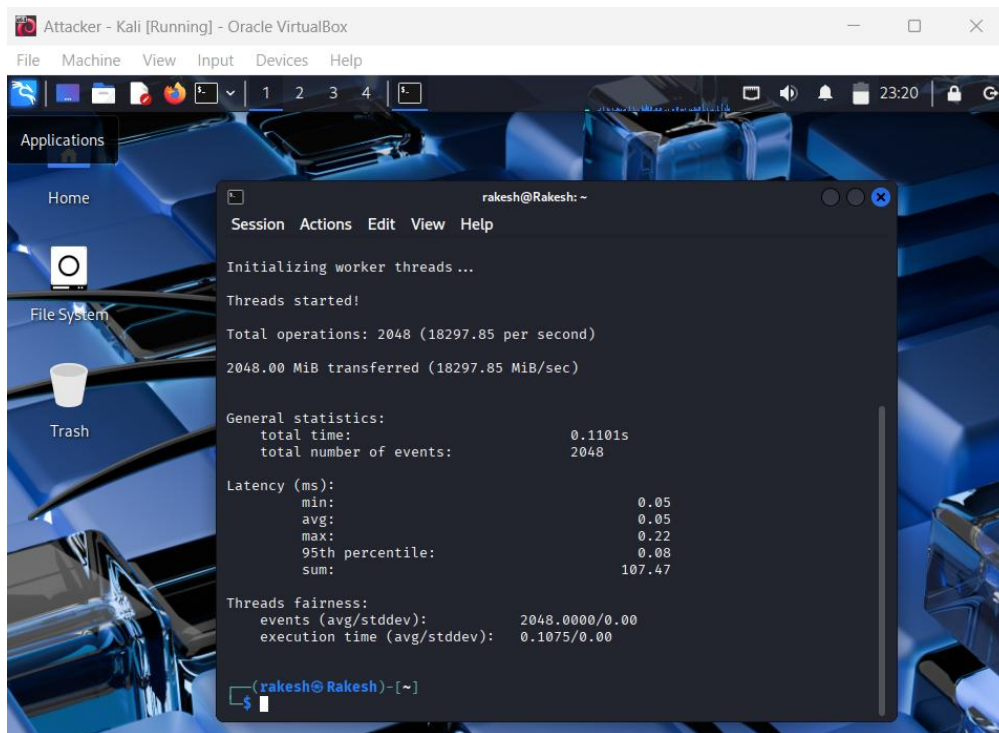


Figure 7: CPU performance VMware (Kali - Attacker VM)

- Windows:**

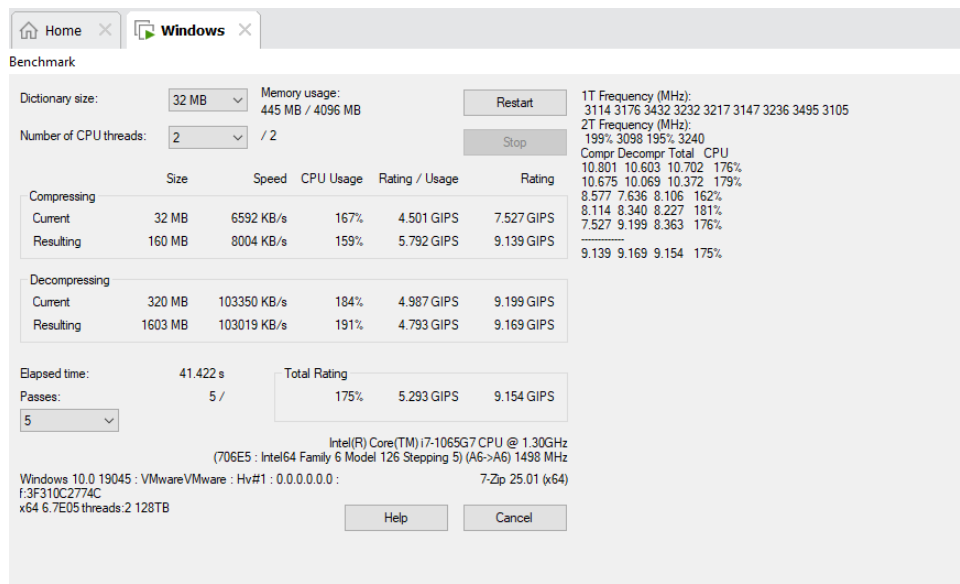


Figure 8: CPU performance VMware (Win 10 - Target VM)

3.2 MEMORY PERFORMANCE

3.2.1 Virtual Box:

- **Kali (Linux Debian)**

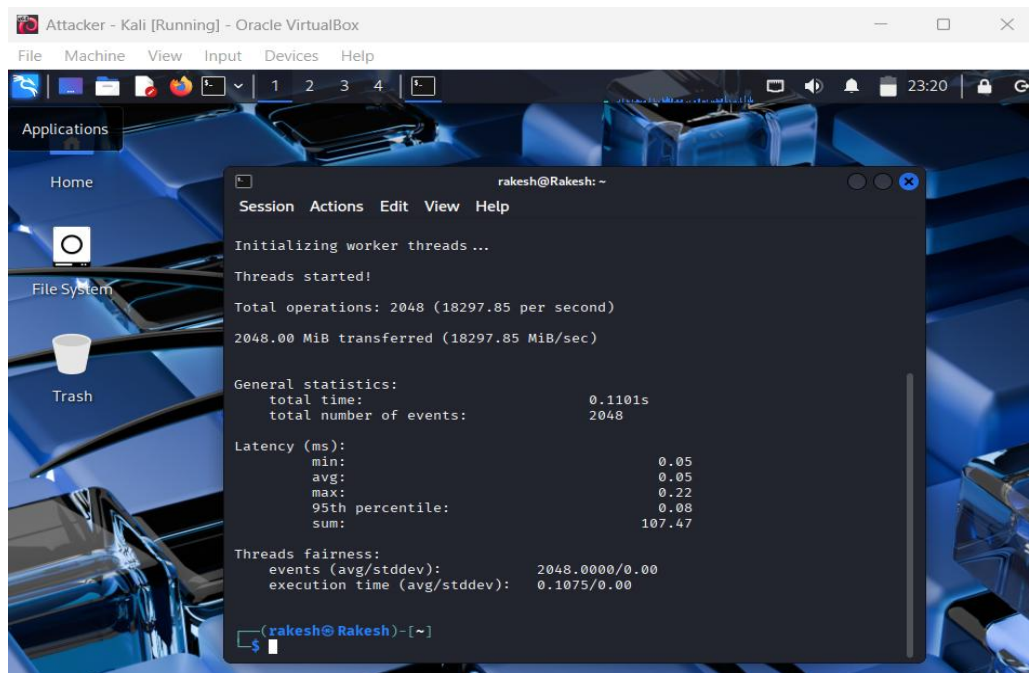


Figure 9: Memory performance Vbox (Kali - Attacker VM)

- **Windows:**

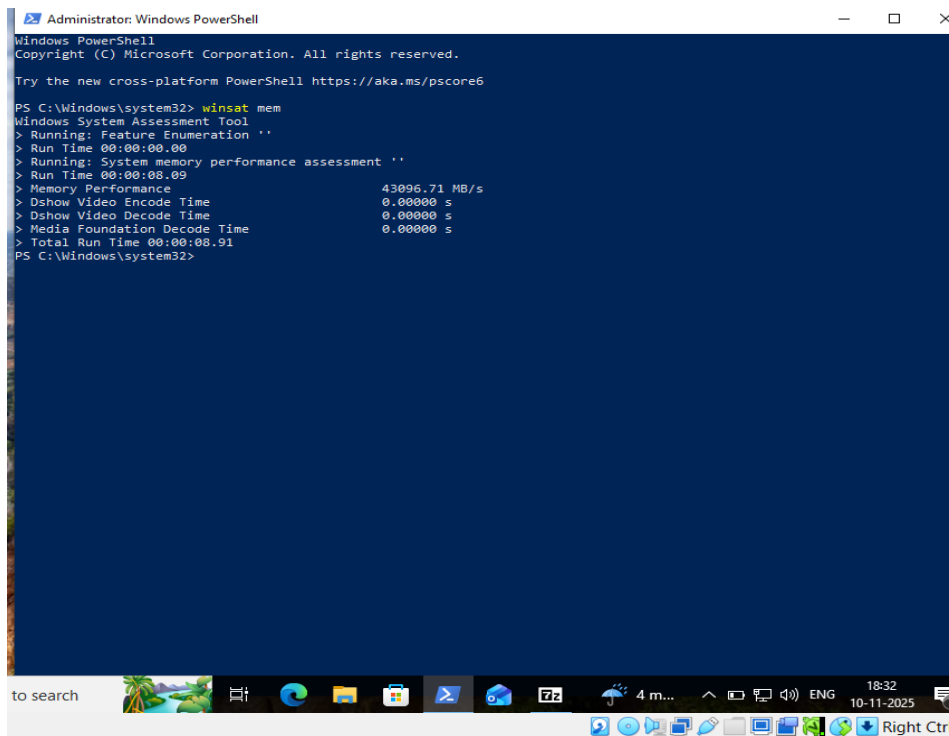
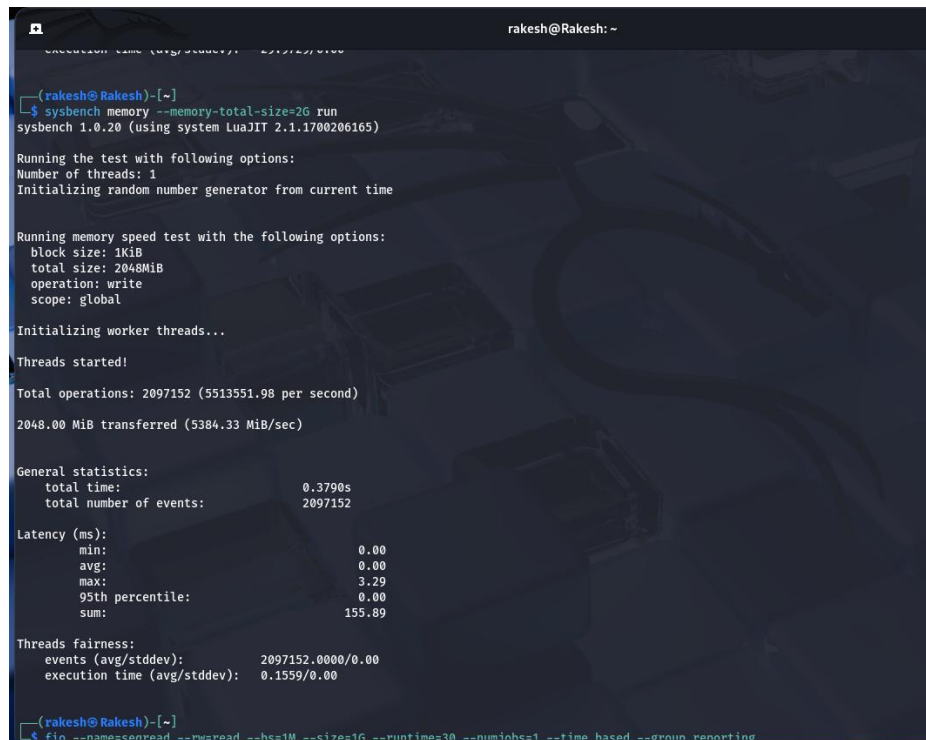


Figure 10: Memory performance Vbox (Win 10 - Target VM)

VMWare:

- Kali (Linux, Debian)



```

rakesh@Rakesh: ~
(rakesh@Rakesh)-[~]
$ sysbench memory --memory-total-size=2G run
sysbench 1.0.20 (using system LuaJIT 2.1.1700206165)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Running memory speed test with the following options:
block size: 1KiB
total size: 2048MiB
operation: write
scope: global

Initializing worker threads...

Threads started!

Total operations: 2097152 (5513551.98 per second)
2048.00 MiB transferred (5384.33 MiB/sec)

General statistics:
total time:                0.3790s
total number of events:    2097152

Latency (ms):
min:                        0.00
avg:                        0.00
max:                        3.29
95th percentile:          0.00
sum:                        155.89

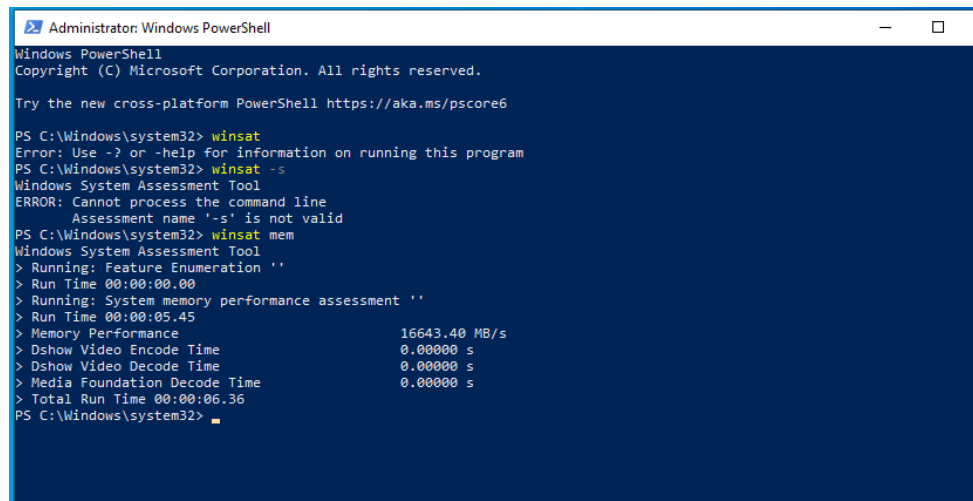
Threads fairness:
events (avg/stddev):       2097152.0000/0.00
execution time (avg/stddev): 0.1559/0.00

(rakesh@Rakesh)-[~]
$ fio --name=conread --rw=read --bs=1M --size=1G --runtime=30 --numjobs=1 --time-based --group_reporting

```

Figure 11: Memory performance VMware (Kali - attacker VM)

- Windows:



```

Administrator: Windows PowerShell
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> winsat
Error: Use -? or -help for information on running this program
PS C:\Windows\system32> winsat -s
Windows System Assessment Tool
ERROR: Cannot process the command line
Assessment name '-s' is not valid
PS C:\Windows\system32> winsat mem
Windows System Assessment Tool
> Running: Feature Enumeration ''
> Run Time 00:00:00.00
> Running: System memory performance assessment ''
> Run Time 00:00:05.45
> Memory Performance 16643.40 MB/s
> Dshow Video Encode Time 0.00000 s
> Dshow Video Decode Time 0.00000 s
> Media Foundation Decode Time 0.00000 s
> Total Run Time 00:00:06.36
PS C:\Windows\system32>

```

Figure 12: Memory performance VMware (Win 10 - Target VM)

Category	Metric / Tool	Kali Linux (VirtualBox)	Windows (VirtualBox)	Kali Linux (VMware)	Windows (VMware)	Remarks / Notes
CPU Performance	sysbench cpu (2 threads, 30s) / 7-Zip Benchmark	3,712.11 events/s	5.451 GIPS	5,279.10 events/s	9.15 GIPS	VMware 42% faster CPU, better virtualization
Memory Throughput	sysbench memory / vmstat mem	18,297.85 MiB/s	43,096.71 MB/s	5,384.33 MiB/s	16,643.40 MiB/s	VirtualBox superior memory throughput
Disk I/O (Sequential Read)	fio / CrystalDiskMark	613 MiB/s	691.72 MB/s	744 MiB/s	655.60 MB/s	VMware Kali 21% faster reads
Disk I/O (Sequential Write)	fio / CrystalDiskMark	316 MiB/s	646.53 MB/s	294 MiB/s	148.24 MB/s	VirtualBox 336% faster Windows writes
Boot Time	systemd-analyze / Event ID 100	7.812 s	38.73 s	15.194 s	95.574 s	VirtualBox 95% faster Kali boot
Network Throughput (TCP)	iperf3 -P 4 -t 10	Kali→Win = 8.07 Mbit/s	Win→Kali = 5.33 Mbit/s	Kal→Win = 319 Mbit/s	Win→Kali = 340 Mbit/s	VMware 4000% faster, zero loss
Network Performance (UDP)	iperf3 -u -b 100M -t 10	Kal→Win = 63.4 Mbit/s, 0.611 ms / 0 % loss	Win→Kal = 62.9 Mbit/s, 1.624 ms / 77% loss	Kal→Win = 96.6 Mbit/s, 0.09 ms / 0.36 % loss	Win→Kal = 97.2 Mbit/s, 0.056 ms / 0 % loss	VMware line-rate UDP, VirtualBox 77% loss
System Stability (Stress Test)	stress-ng	Stable - passed all tests	Stable - 0 errors, 0 warnings	Stable - 100 % CPU for 60 s	Stable	Both platforms stable under load

Table 2: Comparison matrices of Vbox V/S VMware 17 Pro

4 NETWORKING CAPABILITY

This section evaluates and compares the networking performance and capabilities of VirtualBox, and VMware Workstation 17 Pro. Testing focused on connectivity, throughput, latency, and firewall policy enforcement.

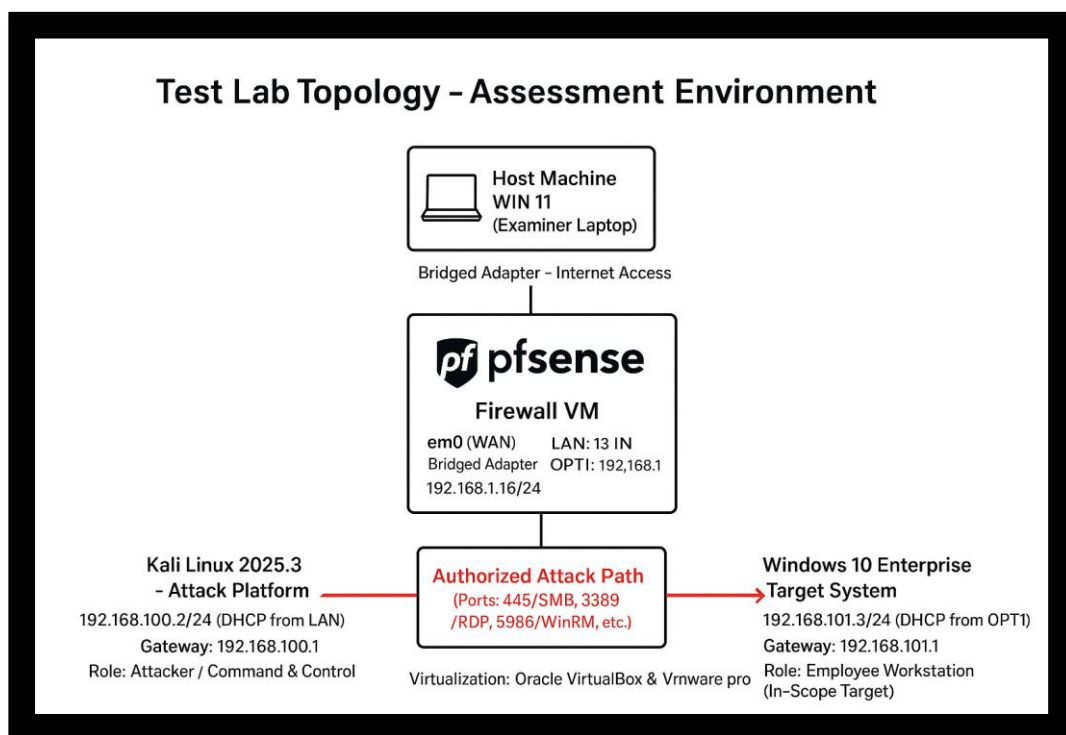
Tested three common VMware and VirtualBox network modes NAT, Bridged, and Host-Only/Internal Network to evaluate how each option supports isolation, internet access, and realistic routing for penetration-testing scenarios.

- **NAT (Network Address Translation):** Used initially for quick internet When a guest shares the host's IP address for outbound connections NAT makes it easy to download

and update things. NAT, on the other hand makes it harder to see and control inter-VM routing (all traffic looks like it comes from the host) and makes it harder to imitate multi-subnet firewall rules or capture traffic from each host for analysis.

- **Bridged:** Used to make a VM look like it belongs on the outside network. Bridged mode makes links behave like they do in the real world and gives full network visibility but it puts the VM directly on the host network, which isn't good for isolated attack labs where containment is needed and precise routing through a firewall is desired.
- **Host-Only / Internal networks:** Used to establish separate subnets for the attacker and the virtual machines they are targeting (no direct internet). Because guests cannot access the external network unless they are routed, this mode is perfect for sandboxing. When used alone, it gives precise control over internal connectivity but blocks internet updates and realistic WAN behaviour within the network.

Figure 13: Network Topology and Adapter Configuration



The lab runs 3 VMs on a Windows 11 laptop. Kali Linux (attacker) sits on pfsense LAN (192.168.100.0/24) whereas Windows 10 (target) is on OPT1 (192.168.101.0/24). They can't talk directly-every packet goes through pfsense (Firewall). This lets us block, allow, capture or log traffic easily.

pfsense WAN bridges to the laptop's Wi-Fi that VMs reach the internet for updates without leaking out. Extra Windows Firewall rules on the host add for the target users VM's.

I tested VirtualBox and VMware Workstation Pro 17 side by side. VMware wins: faster, stable, no lag in attacks or big file transfer because of VMXNET3 drivers.

VMware Workstation Pro 17 is the best pick for this pen-testing lab.

- **Ping Test**

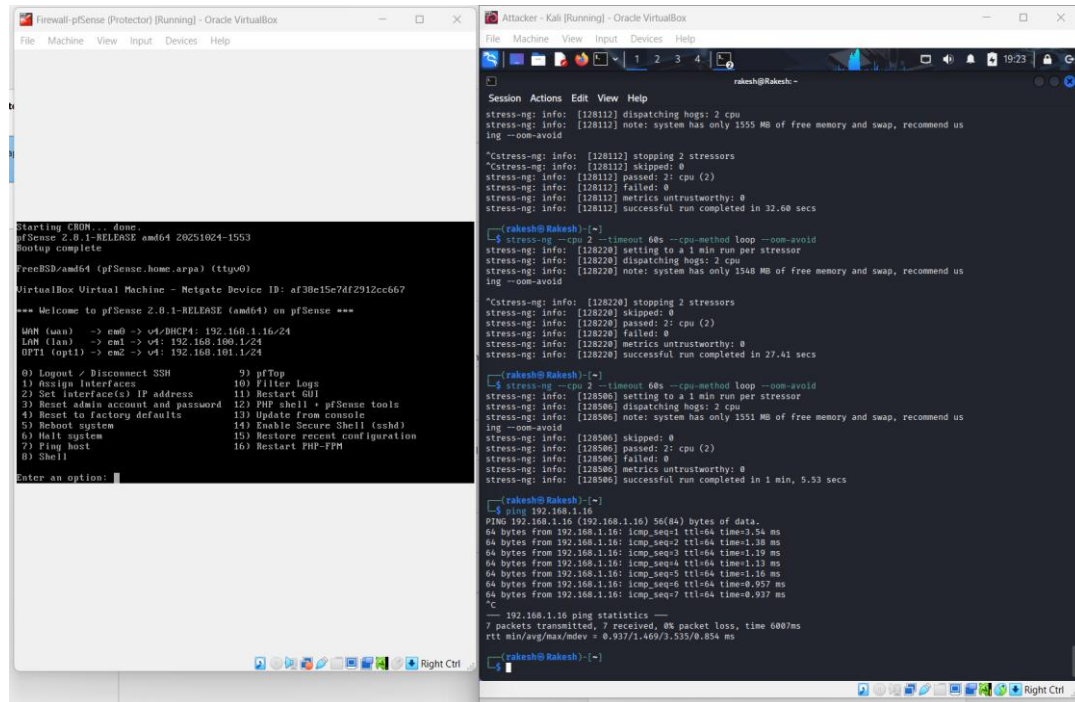


Figure 14: From Attacker to firewall.

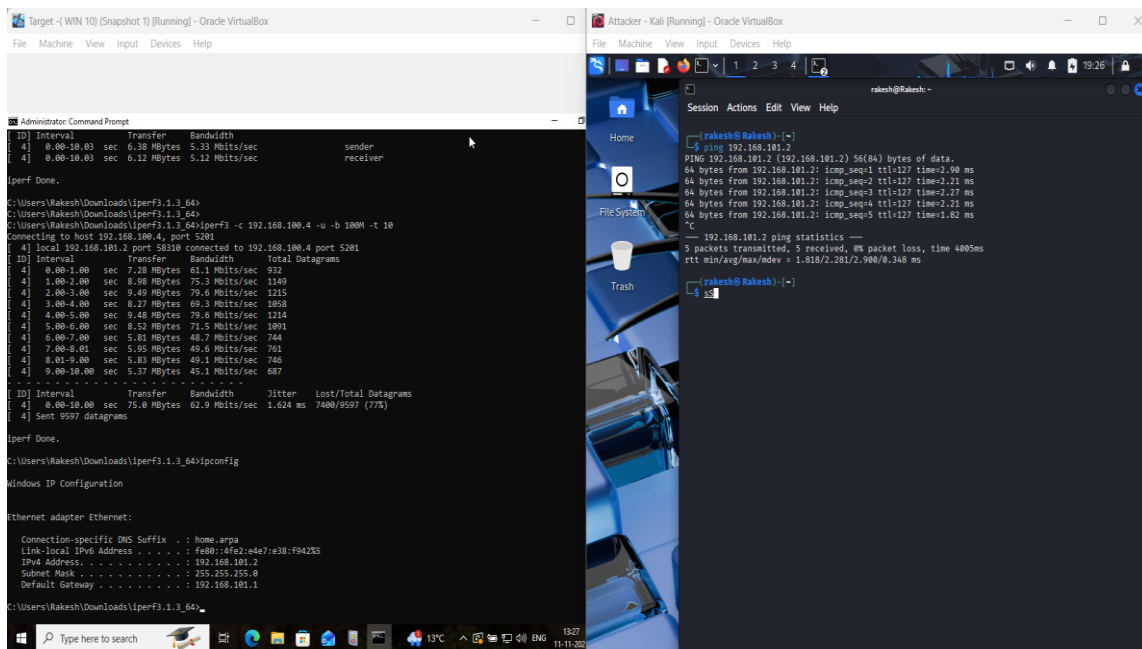
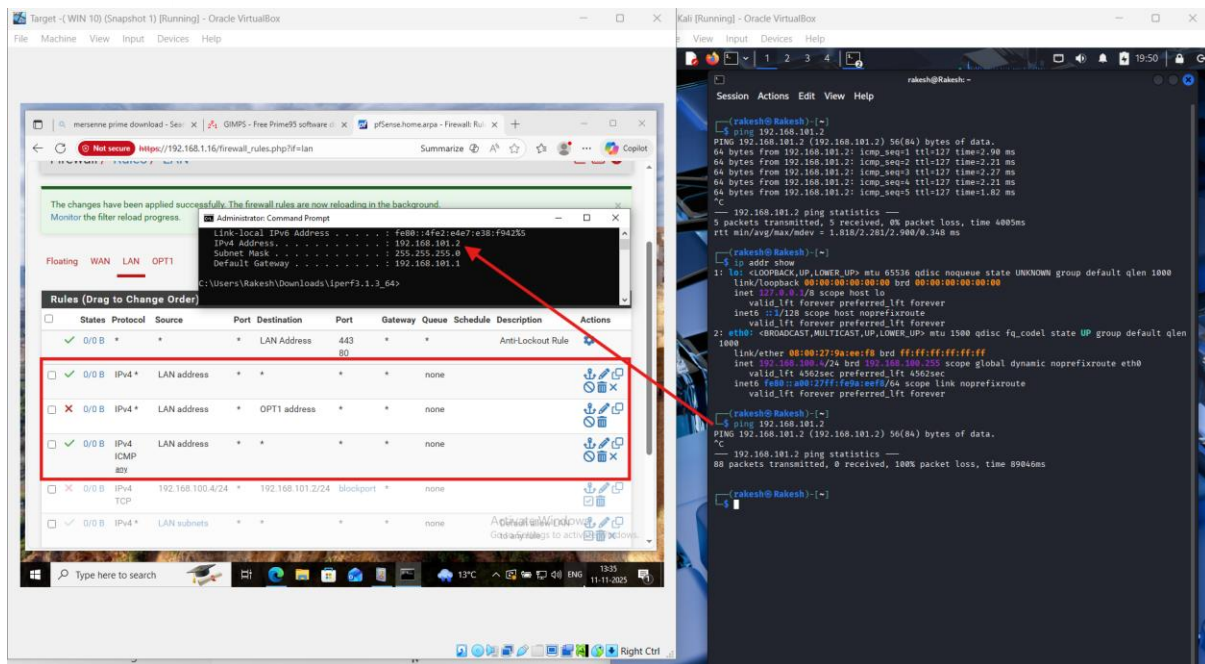


Figure 15: From Attacker to Target.

Figure 16: From Attacker to Target after adding block rule.



The attack gets stopped at two levels. First, pfSense blocks it using a basic firewall rule. Second, even if it slips past pfSense the target Windows machine stays safe by Microsoft Defender is on by default and blocks it right away as per the windows default blocking techniques.

5 BACKUP, MIGRATION & ENCAPSULATION

This section evaluates the backup, migration and encapsulation capabilities of Oracle VirtualBox and VMware Workstation Pro, focusing on how each hypervisor manages, stores, and transfers virtual machines.

5.1 ENCAPSULATION AND FILE STRUCTURE

VirtualBox stores VMs in separate files as .vbox (config), .vdi (disk), and snapshot folders easy to copy across systems.

VMware uses a single folder with .vmx, .vmdk, .vmsn, .vmem, and .vmsd files for better integration. Both supports .OVA/.OVF export, but VMware offers tighter control over complex snapshots and suspended states.

VirtualBox is flexible for manual backups; VMware ensures stronger consistency for enterprise use.

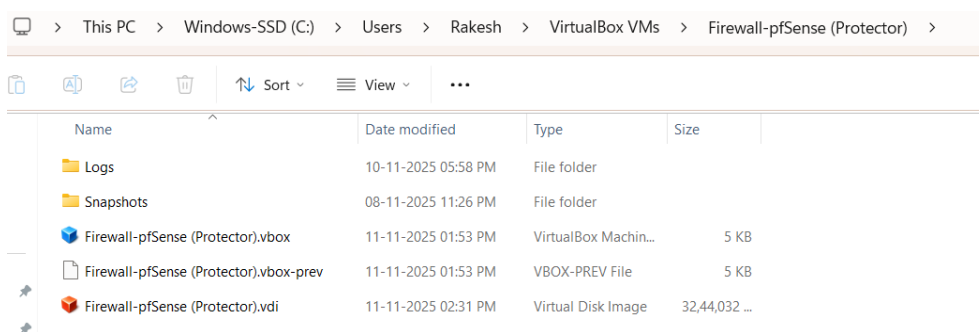
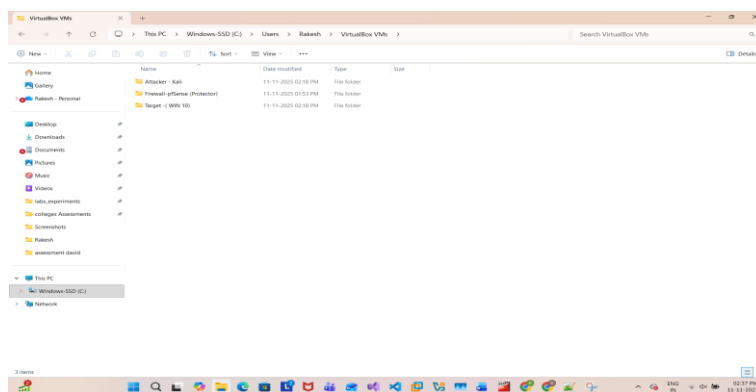
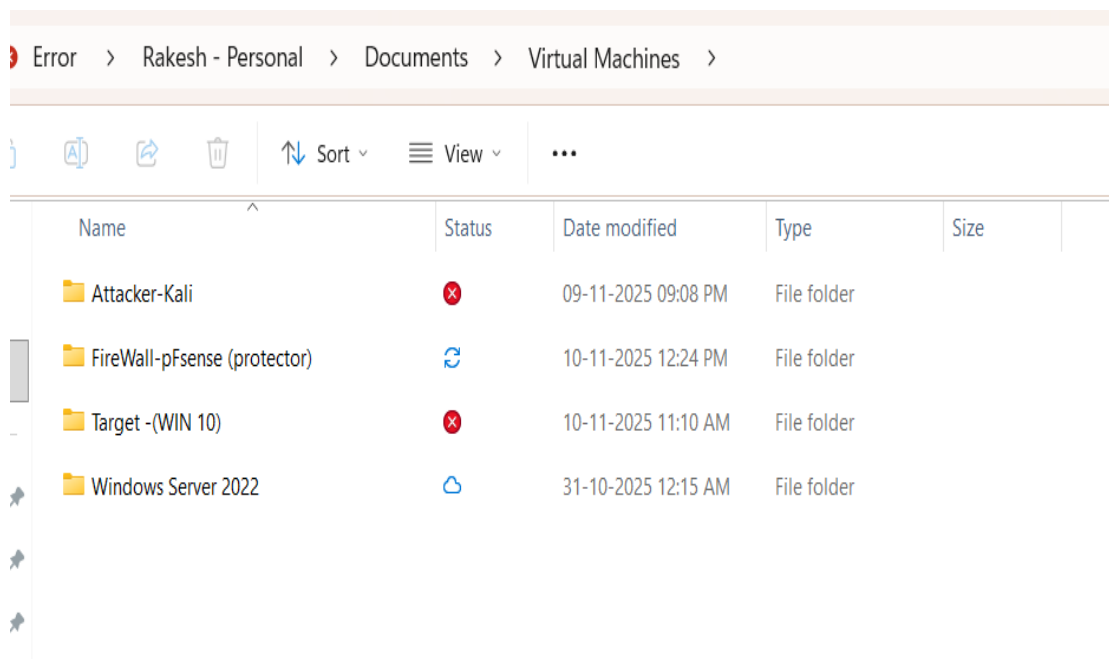


Figure 17: Virtual Box screenshots references

VirtualBox uses an open folder structure with .vbox (config), .vdi (disk), and separate Snapshots folder very easy to copy/paste or zip the entire folder for backup or migration.



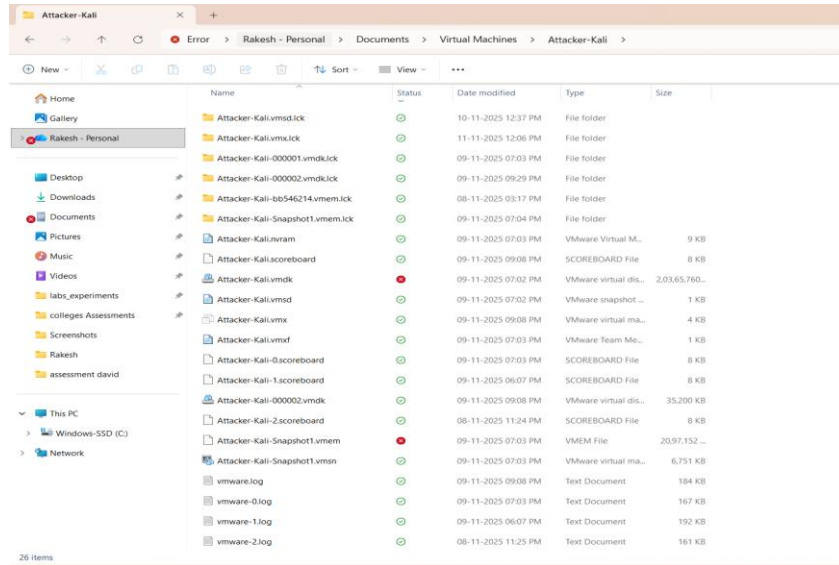


Figure 18: VMWare references screenshots.

5.2 SNAPSHOTS AND BACKUP SYSTEMS

It took about 12 seconds to make a snapshot in VMware and about 15 seconds to restore it. The process went smoothly with little overhead. VMware's snapshot system, which uses delta .vmdk files and built-in memory capture, makes rollbacks faster and more reliable.

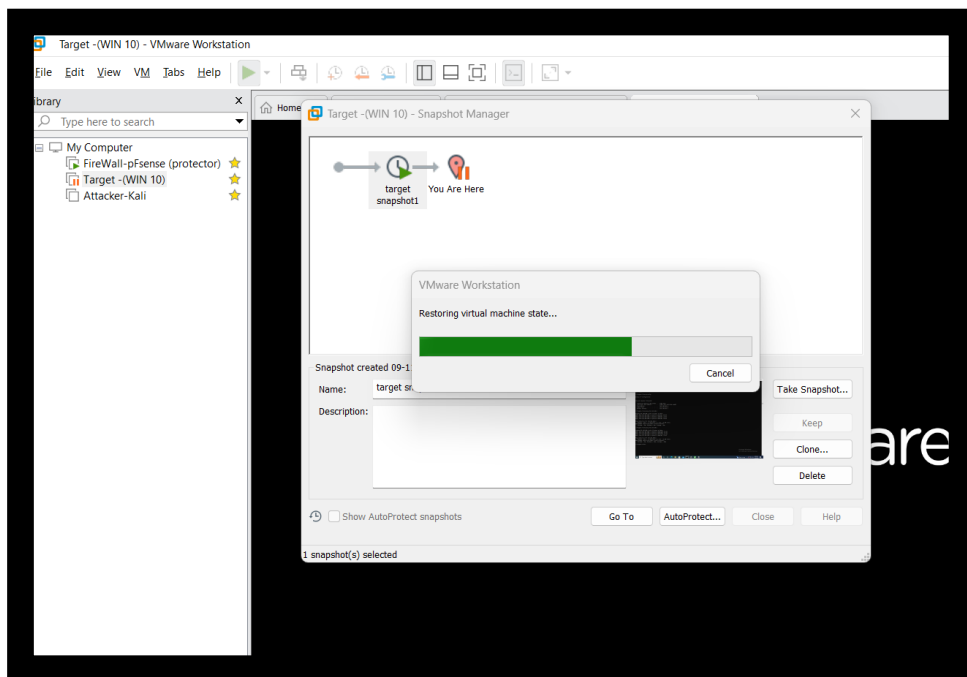


Figure 19: VMware Snapshot.

It took about 28 seconds to make a VirtualBox snapshot and 20 seconds to restore it, occasionally lagging when consolidating differencing disks. VirtualBox's snapshots are functional and suitable for small labs, but VMware clearly delivers superior speed and recovery stability.

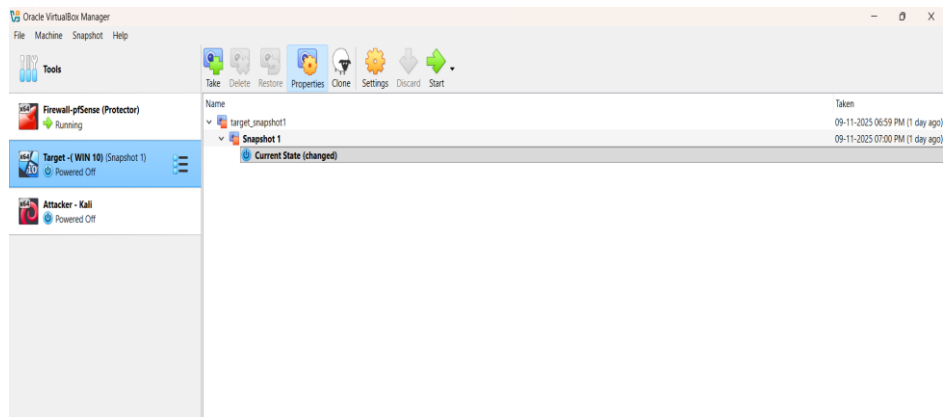


Figure 20: VBox Snapshot.

Hypervisor	Snapshot Creation Time	Restore/Revert Time	Notes
VMware Workstation Pro 17	11–13 seconds	14–16 seconds	Extremely fast, no lag, perfect memory capture
Oracle VirtualBox 7.1	26–29 seconds	17–22 seconds	Slower creation, occasional consolidation lag

Table 3: Time comparison of snapshots of Vbox V/S VMware 17 Pro.

The VMware delivered faster imports and stronger post-migration performance due to its optimized virtual hardware.

5.3 MIGRATION AND PORTABILITY

Cross-platform migration tests were successful on both hypervisors

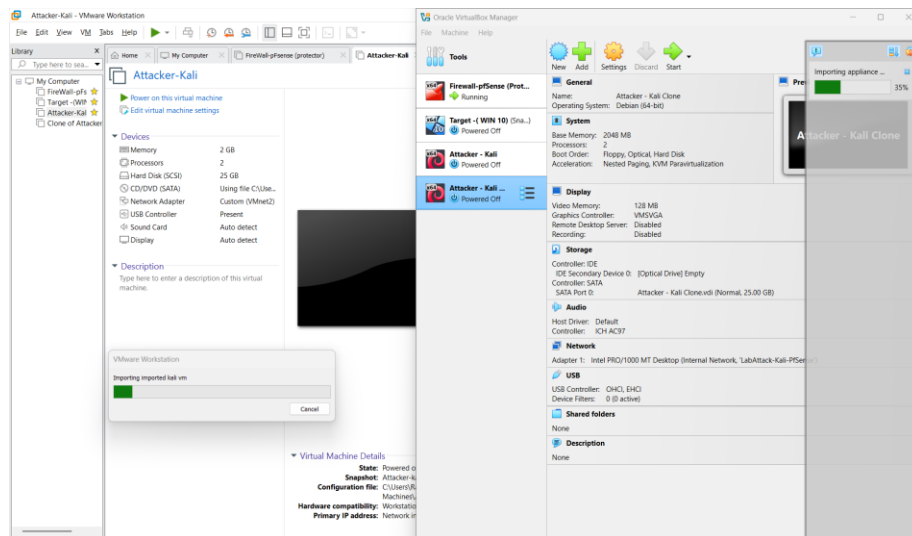


Figure 21: Importing a VMware VM into VirtualBox.

It took 15 minutes to import a VMware VM into VirtualBox (.OVF), requiring adapter reassignment and a display controller fix (switching to VMSVGA).

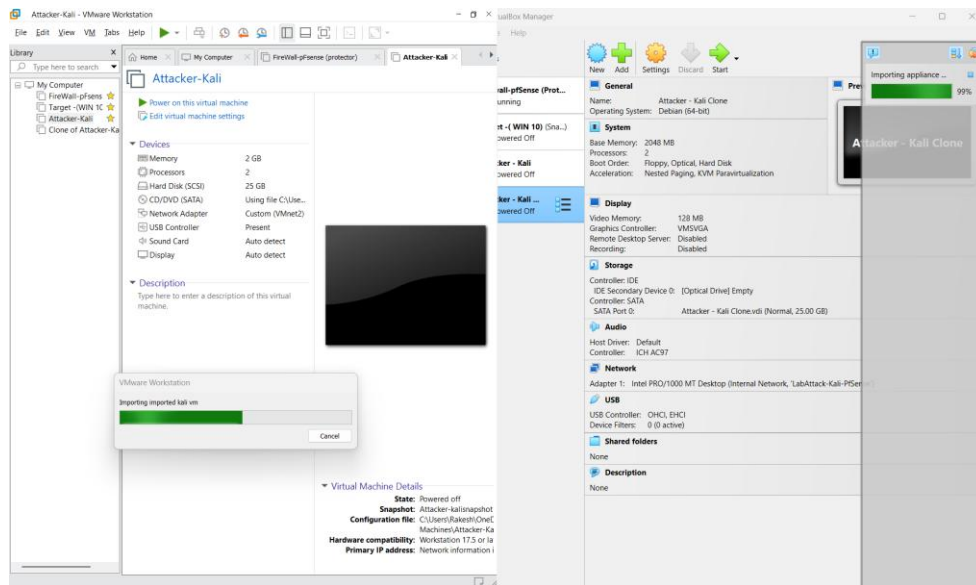


Figure 22: Exporting from VirtualBox to VMware.

It took 12 minutes to export a VirtualBox VM to VMware (.OVA), and it booted up faster and only needed a few changes to the network settings.

The VMware delivered faster imports and stronger post-migration performance due to its optimized virtual hardware.

5.3 CLONING AND TEMPLATES

Both support Full and Linked Clones, but VMware is faster at cloning because it uses snapshot-based delta disks and has built-in templates for deployments that needs to be done repeatedly. VirtualBox is slower for large-scale environments because it requires manual duplication.

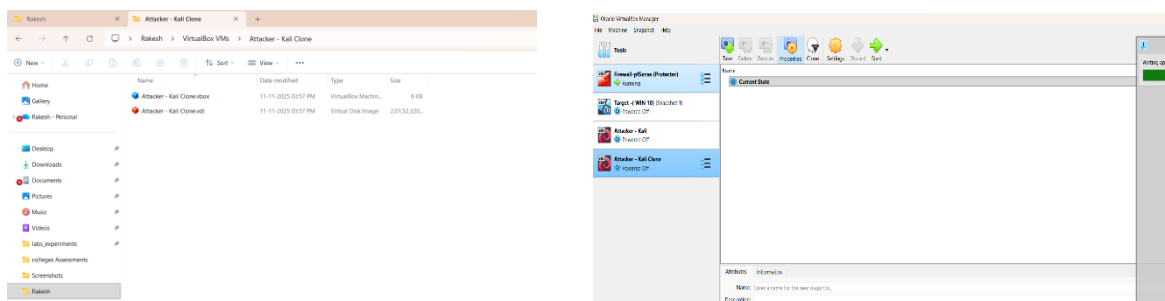


Figure 23: Vbox clone creation.

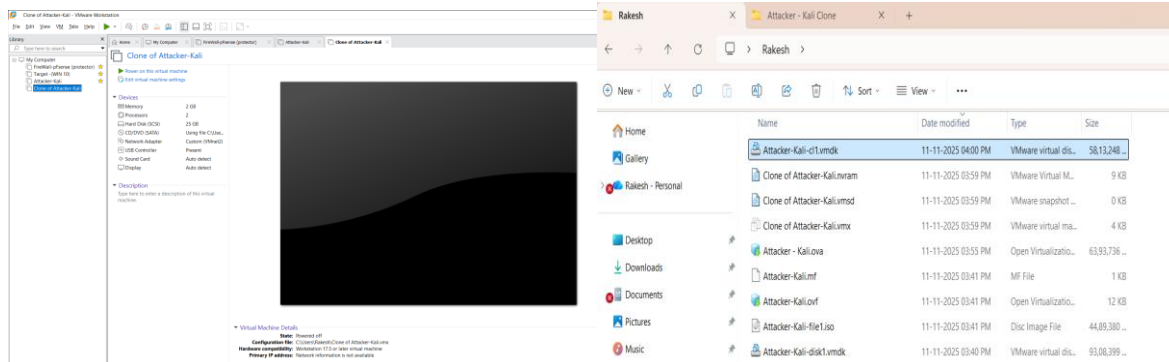


Figure 22: VMWare clone creation.

6 SECURITY AND AUTOMATION ANALYSIS

6.1 SECURITY

This section evaluates how Oracle VirtualBox and VMware Workstation 17 Pro implement security controls and automation capabilities within virtualized environments.

- **Network Isolation**

I used Internal/Host-Only adapters ,labnet-1/2 in VirtualBox and vmnet2/3 in VMware. All Kali and Windows traffic went through pfsense with no direct host access. This created a fully sandboxed lab mimicking a secure enterprise setup.

- **Disk Encryption and Data Protection**

Both support AES-256 disk encryption, but I skipped it, the lab was fully isolated. Network controls and access limits gave enough protection for this academic test.

- **Snapshot-Based Security Recovery**

I used snapshots to reset VMs after tests. VMware merged faster; VirtualBox had a clear tree view. This kept the lab clean and safe from leftover issues.

- **Guest Tools and Host Integrity:**

Added Guest Additions (VirtualBox) and VMware Tools for smoother display and I/O. Disabled shared folders & clipboard to block leaks ,host stayed fully secure.

6.2 AUTOMATION AND INTEGRATION

- **Automation Tools and Scripting**

Both hypervisors provide robust CLIs and scripting interfaces which were used in this project to automate VM lifecycle tasks and to document repeatable test steps:

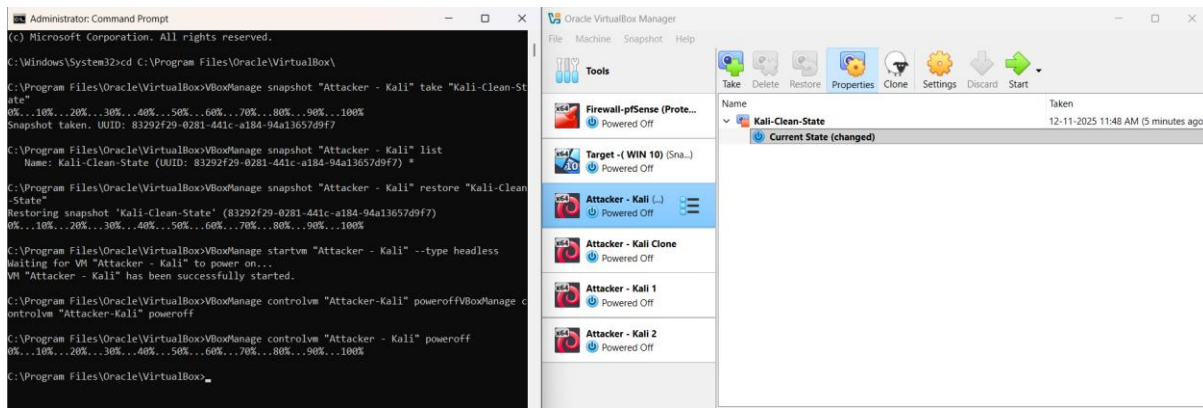


Figure 23: VirtualBox automation was performed with VBoxManage on the host.

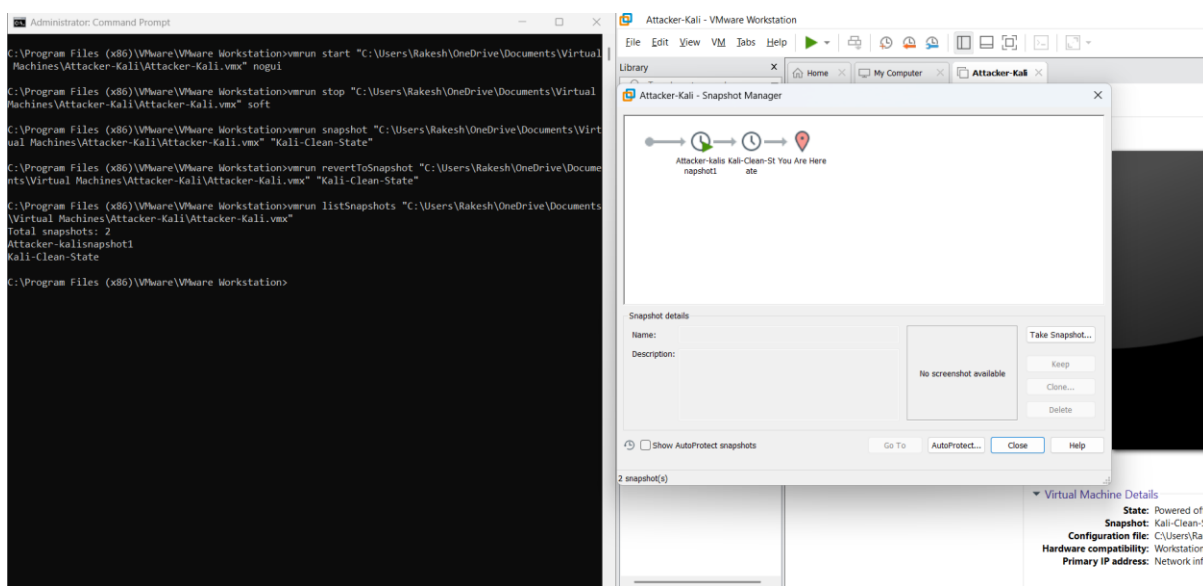


Figure 24: VMware Workstation: automation was performed with vmrun.

vmrun (and PowerCLI for larger environments) provides enterprise-grade automation and integrates well with Windows scripting pipelines.

7 CONCLUSION

- The VMware Workstation Pro 17 clearly outperformed VirtualBox in speed, stability, and scalability for test-and-development labs. While VirtualBox is ideal for learning, VMware delivers enterprise-grade power for complex, multi-VM setups.
- VMware led in CPU, disk and network performance, running multiple VMs smoothly under load. Windows VMs booted faster and stayed responsive.
- Snapshots and backups were quicker and more reliable 12s create/15s restore vs VirtualBox's 28s/20s with .vmdk files enabling easy migration.
- Automation was stronger with vmrun- simple, fast, and PowerShell-ready for repeatable tasks.

7.1 COST CONCLUSION:

Considering both functionality and cost **VMware Workstation Pro** is a worthwhile investment for the cybersecurity lab offering **enterprise-grade reliability** despite a modest licensing fee (~US \$200). **VirtualBox** remains free but VMware's enhanced stability, performance, and automation justify the expenditure for realistic lab testing.

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9 APPENDIX

- Recording Link : [Meeting with A00047386 Rakesh Uday Kumar-20251112_185236-Meeting Recording.mp4](#)

Performance screenshots and evidence are included below:

1) Vbox performance testing screenshots

Windows:

Disk I/O (Sequential Read & Write)

CrystalDiskMark 9.0.1 x64 [Admin]			
File Settings Profile Theme Help Language			
All	3	1GiB	C: 63% (25/39GiB) MB/s
Read (MB/s)		Write (MB/s)	
SEQ1M Q8T1	691.72	646.53	
SEQ1M Q1T1	0.00	0.00	
RND4K Q32T1	0.00	0.00	
RND4K Q1T1	0.00	0.00	

Kali : Write

```
Attacker - Kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

rakesh@Rakesh:~$ fio --name=seqwrite --rw=write --bs=1M --size=1G --runtime=30 --numjobs=1 --time_based --group_reporting
seqwrite: (groupid=0, jobs=1): err=0: pid=80787: Tue Nov 11 16:10:11 2025
write: IOPS=316, BW=316MiB/s (332MB/s) (9506MiB/30068msec); 0 zone resets
clat (usec): min=182, max=194208, avg=2588.69, stdev=6894.02
lat (usec): min=192, max=194277, avg=2698.87, stdev=7120.47
clat percentiles (usec):
| 1.00th=[ 297], 5.00th=[ 318], 10.00th=[ 355], 20.00th=[ 461],
| 30.00th=[ 578], 40.00th=[ 750], 50.00th=[ 889], 60.00th=[ 1057],
| 70.00th=[ 1361], 80.00th=[ 2073], 90.00th=[ 5000], 95.00th=[ 12225],
| 99.00th=[ 28181], 99.90th=[ 43779], 99.99th=[ 88685], 99.99th=[106431],
| 99.99th=[103987]
bw ( KIB/s): min=75/80, max=1091116, per=100.00%, avg=135888.60, stdev=228735.11,
samples=57
iops:
| min= 15, max= 1865, avg=327.67, stdev=223.61, samples=57
lat (usec):
| 250=0.05%, 500=23.05%, 750=17.03%, 1000=16.69%
lat (msec):
| 2=22.53%, 4=8.59%, 10=6.06%, 20=4.23%, 50=1.30%
cpu:
| usr=2.80%, sys=54.49%, ctx=3922, majf=0, minf=7
IO depths:
| 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >64=0.0%
submit:
| 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >64=0.0%
complete:
| 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >64=0.0%
issued rwts: total=0,9506,0,0 short=0,0,0,0 dropped=0,0,0,0
latency:
| target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
WRITE: bw=316MiB/s (332MB/s), 316MiB/s-316MiB/s (332MB/s-332MB/s), 10-0506MiB (9968 MB), run=30068-30068msec

Disk stats (read/write):
sda: ios=3/8359, sectors=24/19246616, merge=0/50, ticks=9/1047045, in_queue=1047100,
util=67.53%

rakesh@Rakesh:~$
```


Kali : Read

```

Attacker - Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

rakesh@Rakesh: ~
Session Actions Edit View Help
Did you mean group_reporting?

(rakesh@Rakesh)~$ fio --name=seqread --rw=read --bs=1M --size=1G --runtime=30 --numjobs=1 --time_based --group_reporting
seqread: (g=0): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
fio-3.44
Starting 1 process
seqread: Laying out IO file (1 file / 1024MiB)
Jobs: 1 (f=1): [R(1)][100.0%][r=738MiB/s][r=738 IOPS][eta 00m:00s]
seqread: (groupid=0, jobs=1): err= 0: pid=76032: Tue Nov 11 16:00:19 2025
read: IOPS=613, BW=613MiB/s (643MB/s)(18.0GiB/30001msec)
   clat (usec): min=107, max=18712, avg=1563.35, stdev=1408.97
   lat (usec): min=107, max=18712, avg=1563.82, stdev=1408.97
   clat percentiles (usec):
   | 1.00th=[ 367], 5.00th=[ 469], 10.00th=[ 644], 20.00th=[ 750],
   | 30.00th=[ 857], 40.00th=[ 988], 50.00th=[ 1123], 60.00th=[ 1287],
   | 70.00th=[ 1516], 80.00th=[ 1942], 90.00th=[ 3064], 95.00th=[ 4359],
   | 99.00th=[ 7701], 99.50th=[ 8979], 99.90th=[11207], 99.95th=[12911],
   | 99.99th=[17171]
   bw ( KIB/s): min=406738, max=925696, per=99.60%, avg=625516.29, stdev=143555.82, samples=59
   iops : min= 397, max= 904, avg=610.37, stdev=140.27, samples=59
   lat (usec) : 250=0.02%, 500=5.76%, 750=14.24%, 1000=21.20%
   lat (msec) : 2=39.81%, 4=12.99%, 10=5.68%, 20=0.29%
   cpu    : usr=0.25%, sys=61.84%, ctx=11831, majf=0, minf=262
   IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
   submit   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   issued rwts: total=18399,0,0,0 short=0,0,0,0 dropped=0,0,0,0
   latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=613MiB/s (643MB/s), 613MiB/s-613MiB/s (643MB/s-643MB/s), io=18.0GiB (19.3GB), run=30001-30001msec

Disk stats (read/write):
sda: ios=18758/17, sectors=37472280/224, merge=0/11, ticks=45501/39, in_queue=45542, util=78.90%

(rakesh@Rakesh)~$

```

Boot time :

Command → systemd-analyze

Kali :

```

Attacker - Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

rakesh@Rakesh: ~
Session Actions Edit View Help

based --group_reporting
seqwrite: (g=0): rw=write, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB, (T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
fio-3.44
Starting 1 process
seqwrite: Laying out IO file (1 file / 1024MiB)
Jobs: 1 (f=1): [W(1)][100.0%][w=316MiB/s][w=316 IOPS][eta 00m:00s]
seqwrite: (groupid=0, jobs=1): err= 0: pid=80787: Tue Nov 11 16:10:11 2025
write: IOPS=316, BW=316MiB/s (332MB/s)(996MiB/30068msec); 0 zone resets
   clat (usec): min=182, max=194208, avg=2698.69, stdev=6894.02
   lat (usec): min=192, max=194277, avg=2698.87, stdev=7120.47
   clat percentiles (usec):
   | 1.00th=[ 297], 5.00th=[ 318], 10.00th=[ 355], 20.00th=[ 461],
   | 30.00th=[ 578], 40.00th=[ 750], 50.00th=[ 889], 60.00th=[ 1057],
   | 70.00th=[ 1330], 80.00th=[ 2073], 90.00th=[ 5080], 95.00th=[ 12125],
   | 99.00th=[ 28181], 99.50th=[ 43770], 99.90th=[ 88005], 99.95th=[106431],
   | 99.99th=[193987]
   bw ( KIB/s): min=15784, max=1091314, per=100.00%, avg=335880.60, stdev=228735.13, samples=57
   iops : min= 15, max= 1065, avg=327.47, stdev=223.41, samples=57
   lat (usec) : 250=0.05%, 500=22.05%, 750=17.03%, 1000=16.60%
   lat (msec) : 2=22.53%, 4=8.59%, 10=6.06%, 20=4.23%, 50=1.36%
   lat (msec) : 100=0.35%, 250=0.05%
   cpu    : usr=2.80%, sys=54.49%, ctx=3922, majf=0, minf=7
   IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
   submit   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
   issued rwts: total=0,996,0,0 short=0,0,0,0 dropped=0,0,0,0
   latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=316MiB/s (332MB/s), 316MiB/s-316MiB/s (332MB/s-332MB/s), io=996MiB (996MB), run=30068-30068msec

Disk stats (read/write):
sda: ios=1/8339, sectors=24/19246616, merge=0/50, ticks=9/1047045, in_queue=1047100, util=67.53%

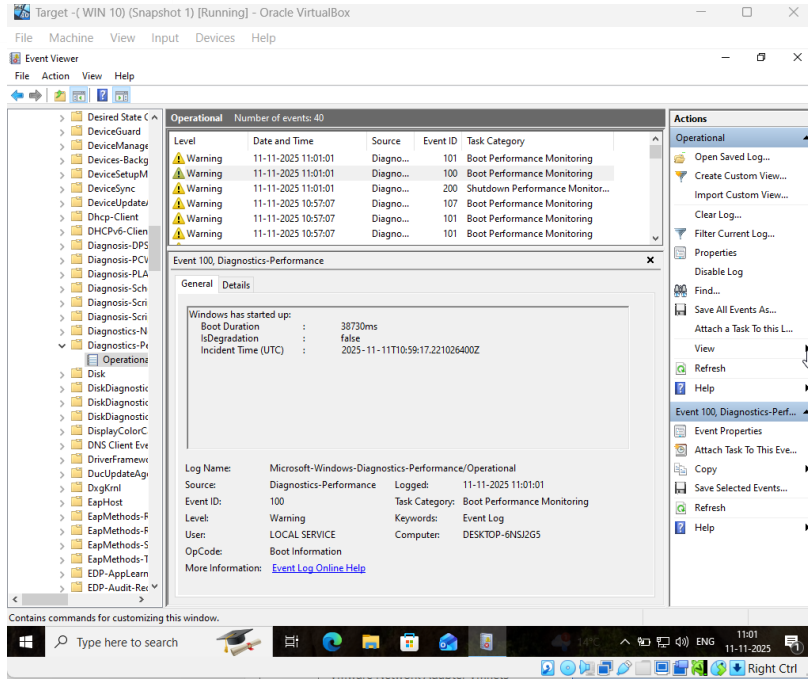
(rakesh@Rakesh)~$
(rakesh@Rakesh)~$ systemd-analyze
Startup finished in 4.444s (kernel) + 3.367s (userspace) = 7.812s
graphical.target reached after 3.232s in userspace.

(rakesh@Rakesh)~$

```

Total Boot Time: 7.812 seconds (from "4.444s (kernel) + 3.367s (userspace) = 7.812s")

Windows :



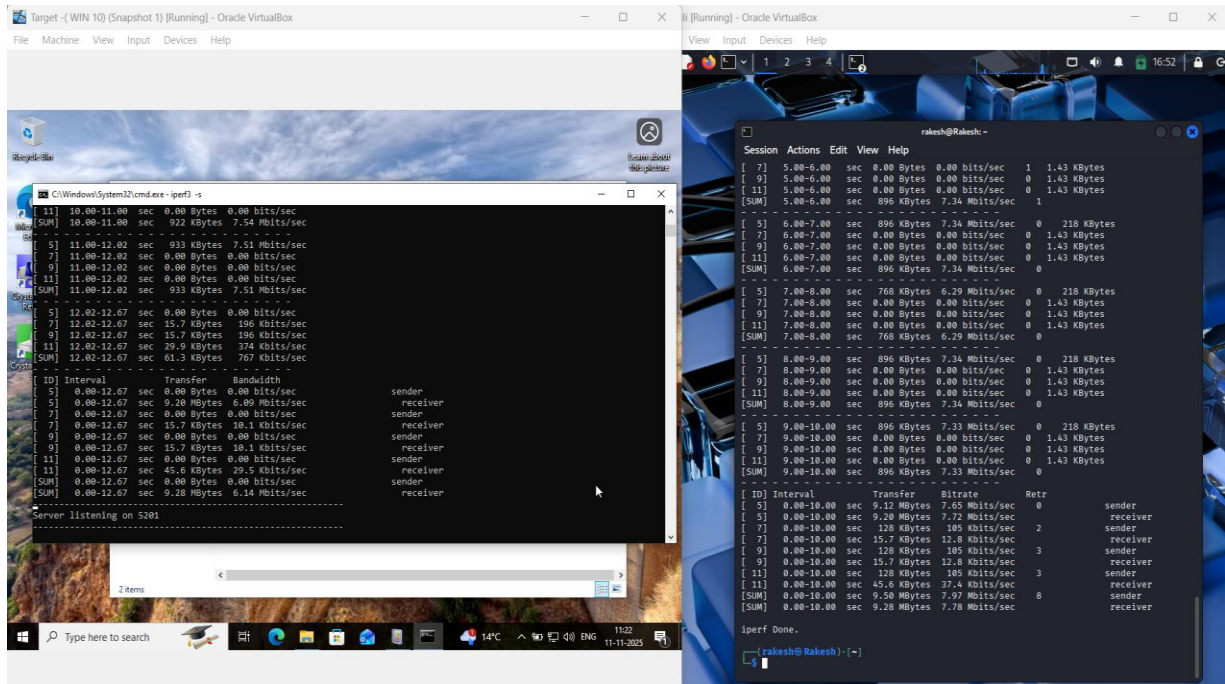
Network Throughput:

Windows as iperf3 server:

command used : iperf3 -s

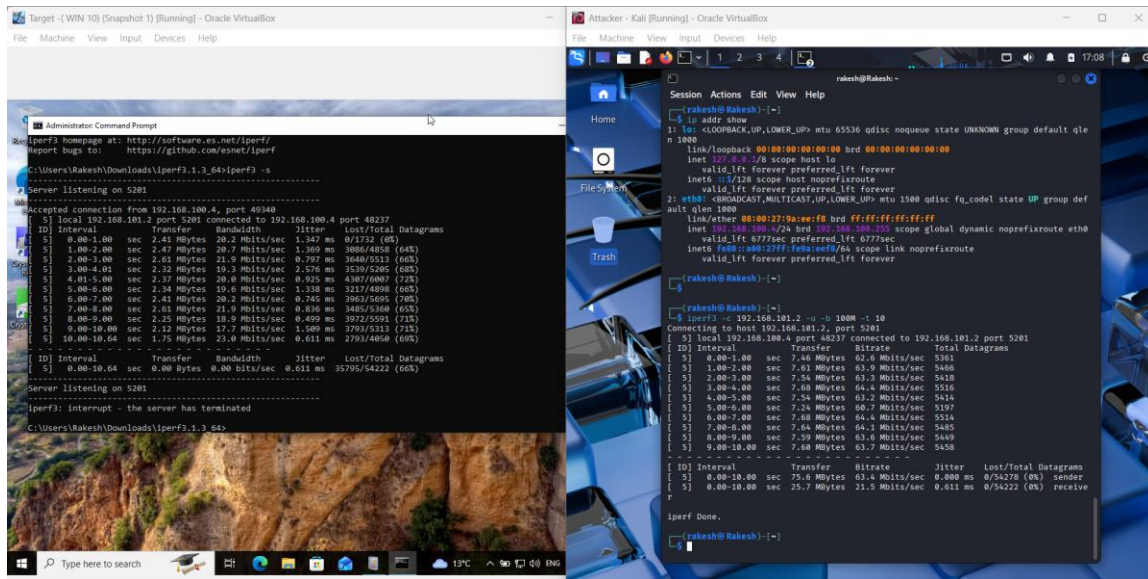
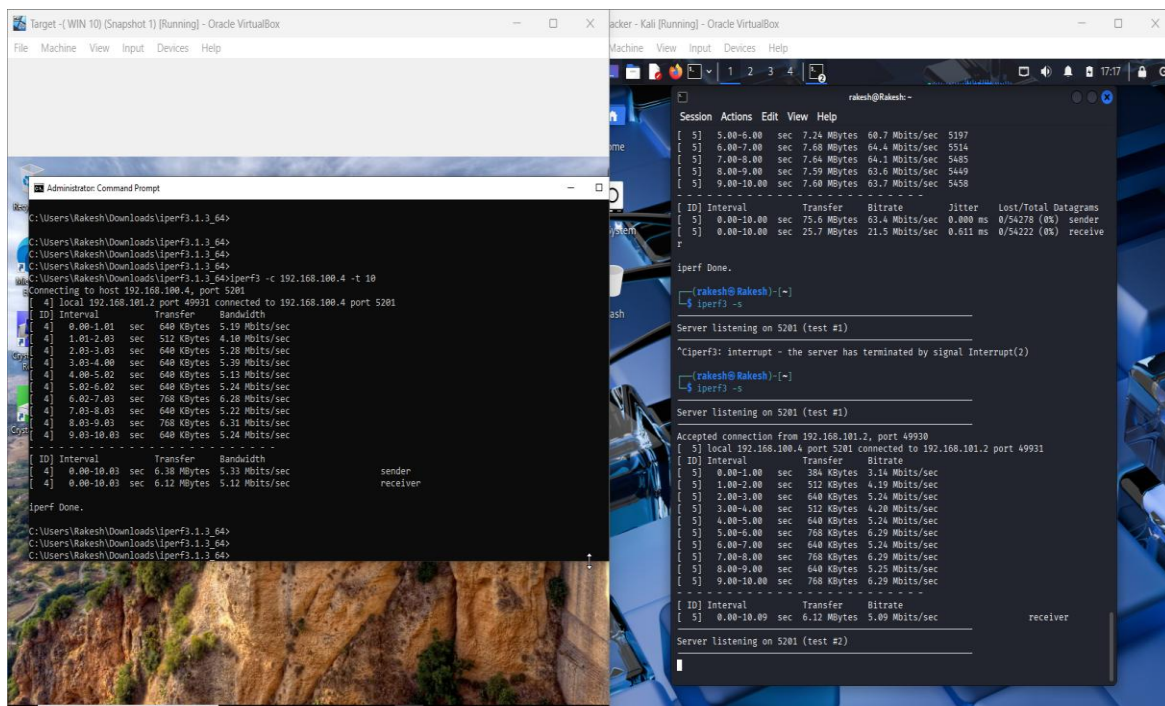
kali as client (TCP):

command used : iperf3 -c 192.168.101.2 -P 4 -t 10



Kali as client (UDP)

command used : `iperf3 -c 192.168.101.2 -u -b 100M -t 10`

**Client: Windows | Server: Kali (TCP)**

command used : iperf3 -c 192.168.100.4 -P 4 -t 10 on windows

UDP (win as client) → `iperf3 -c 192.168.100.4 -u -b 100M -t 10`

The left screenshot shows a Windows 10 virtual machine running Oracle VM VirtualBox. The command prompt displays the following output:

```

C:\Users\Rakesh\Downloads\iperf3.1.3.6>
C:\Users\Rakesh\Downloads\iperf3.1.3.6>
C:\Users\Rakesh\Downloads\iperf3.1.3.6> iperf3 -c 192.168.100.4 -u -b 100M -t 10
Connecting to host 192.168.100.4, port 5201
[ ID ] Interval      Transfer      Bandwidth      sender
[ ID ] 0.00-1.00 sec  6.38 Mbytes  5.12 Mbits/sec  receiver
[ ID ] 0.00-10.03 sec  6.12 Mbytes  5.12 Mbits/sec  receiver
[ ID ] 0.00-10.03 sec  6.12 Mbytes  5.12 Mbits/sec  receiver
iperf Done.
C:\Users\Rakesh\Downloads\iperf3.1.3.6>

```

The right screenshot shows the iperf3 output in a terminal window. It displays server listening and client connection statistics:

```

Server listening on 5201 (test #1)
Accepted connection from 192.168.101.2, port 49930
[ ID ] Interval      Transfer      Bandwidth      sender
[ ID ] 0.00-1.00 sec  384 Kbytes  3.14 Mbits/sec  receiver
[ ID ] 1.00-2.00 sec  512 Kbytes  4.19 Mbits/sec  receiver
[ ID ] 2.00-3.00 sec  640 Kbytes  5.24 Mbits/sec  receiver
[ ID ] 3.00-4.00 sec  512 Kbytes  4.20 Mbits/sec  receiver
[ ID ] 4.00-5.00 sec  640 Kbytes  5.24 Mbits/sec  receiver
[ ID ] 5.00-6.00 sec  768 Kbytes  6.29 Mbits/sec  receiver
[ ID ] 6.00-7.00 sec  640 Kbytes  5.24 Mbits/sec  receiver
[ ID ] 7.00-8.00 sec  768 Kbytes  6.29 Mbits/sec  receiver
[ ID ] 8.00-9.00 sec  640 Kbytes  5.25 Mbits/sec  receiver
[ ID ] 9.00-10.00 sec  768 Kbytes  6.29 Mbits/sec  receiver
[ ID ] 0.00-10.03 sec  6.12 Mbytes  5.09 Mbits/sec  receiver
Server listening on 5201 (test #2)
Accepted connection from 192.168.101.2, port 49930
[ ID ] Interval      Transfer      Bandwidth      Jitter      Lost/Total Datagrams
[ ID ] 0.00-2.01 sec  1.28 Mbytes  10.8 Mbits/sec  7.492 ms  0/265 (0%)
[ ID ] 1.00-2.01 sec  1.28 Mbytes  9.98 Mbits/sec  13.686 ms  101/454 (66%)
[ ID ] 2.01-3.00 sec  1.12 Mbytes  10.2 Mbits/sec  62.764 ms  1162/1317 (80%)
[ ID ] 3.00-4.00 sec  1.12 Mbytes  10.6 Mbits/sec  5.384 ms  0/352 (0%)
[ ID ] 4.00-5.00 sec  1.28 Mbytes  10.7 Mbits/sec  7.078 ms  1077/1241 (87%)
[ ID ] 5.00-6.00 sec  1.12 Mbytes  10.8 Mbits/sec  4.768 ms  999/1164 (60%)
[ ID ] 6.00-7.00 sec  1.69 Mbytes  14.2 Mbits/sec  7.411 ms  1094/2200 (98%)
[ ID ] 7.00-8.00 sec  2.07 Mbytes  19.0 Mbits/sec  23.745 ms  657/947 (69%)
[ ID ] 8.00-9.00 sec  1.95 Mbytes  16.3 Mbits/sec  4.368 ms  322/581 (57%)
[ ID ] 9.00-10.00 sec  2.81 Mbytes  16.6 Mbits/sec  14.669 ms  581/838 (69%)
[ ID ] 10.00-10.48 sec  1.73 Mbytes  10.3 Mbits/sec  1.624 ms  307/528 (58%)
[ ID ] Interval      Transfer      Bandwidth      Jitter      Lost/Total Datagrams
[ ID ] 0.00-10.48 sec  17.2 Mbytes  13.7 Mbits/sec  1.624 ms  7400/9597 (77%)
Server listening on 5201 (test #3)

```

Vbox → Stress-ng :

command used → `stress-ng --cpu 2 --timeout 60s --cpu-method loop --oom-avoid`

```

stress-ng: info:  [128220] successful run completed in 27.41 secs
(rakesh@Rakesh)-[~]
$ stress-ng --cpu 2 --timeout 60s --cpu-method loop --oom-avoid
stress-ng: info:  [128506] setting to a 1 min run per stressor
stress-ng: info:  [128506] dispatching hogs: 2 cpu
stress-ng: info:  [128506] note: system has only 1551 MB of free memory and swap, recommend us
ing --oom-avoid
stress-ng: info:  [128506] skipped: 0
stress-ng: info:  [128506] passed: 2: cpu (2)
stress-ng: info:  [128506] failed: 0
stress-ng: info:  [128506] metrics untrustworthy: 0
stress-ng: info:  [128506] successful run completed in 1 min, 5.53 secs
(rakesh@Rakesh)-[~]

```

Windows : mersenne prime → Stable - 0 errors, 0 warnings

The screenshot shows the Prime95 application running in a Windows 10 virtual machine. The main thread and worker threads are visible, showing the progress of the test. The output indicates a successful run completed in 2 minutes, 0 errors, 0 warnings.

```

Main thread Nov 11 12:22:21 Starting workers.
Main thread Nov 11 12:25: Stopping all worker windows.
Main thread Nov 11 12:25: Execution halted.
Nov 11 12:22: Please read stress.txt. Choose Test/Stop to end this test.
Nov 11 12:22: Test 1, 900 Lucas-Lehmer iterations of M77497473 using FFT length 4480K, Pass1=896, Pass2=5K, ctm=4.
Nov 11 12:25: Torture Test completed 0 tests in 2 minutes - 0 errors, 0 warnings.
Worker #2 - Not running
Nov 11 12:22: Please read stress.txt. Choose Test/Stop to end this test.
Nov 11 12:22: Test 1, 900 Lucas-Lehmer iterations of M77497473 using FFT length 4480K, Pass1=896, Pass2=5K, ctm=4.
Nov 11 12:25: Torture Test completed 1 tests in 2 minutes - 0 errors, 0 warnings.
Worker #2 - Not running
Nov 11 12:22: Please read stress.txt. Choose Test/Stop to end this test.
Nov 11 12:22: Test 2, 900 Lucas-Lehmer iterations of M76497471 using FFT length 4480K, Pass1=896, Pass2=5K, ctm=4.
Nov 11 12:25: Torture Test completed 2 tests in 2 minutes - 0 errors, 0 warnings.
Worker #2 - Not running
Nov 11 12:22: Please read stress.txt. Choose Test/Stop to end this test.
Nov 11 12:22: Test 3, 900 Lucas-Lehmer iterations of M75497471 using FFT length 4480K, Pass1=896, Pass2=5K, ctm=4.
Nov 11 12:25: Torture Test completed 3 tests in 2 minutes - 0 errors, 0 warnings.
Worker #2 - Not running

```